International Journal of Food Science +Technology

International Journal of Food Science and Technology 2015, **50**, 1065–1075

Review

Puff pastry and trends in fat reduction: an update

Kanchana S. Wickramarachchi,[1] Mike J. Sissons[2]* & Stanley P. Cauvain[3,4]

1 FOODplus Research Centre, Waite Research Institute, University of Adelaide, Glen Osmond, SA 5064, Australia
2 NSW Department of Primary Industries, Tamworth Agricultural Institute, 4 Marsden Park Road, Calala, NSW 2340, Australia
3 Bake Tran, 1 Oakland Close, Freeland, Witney, OX 29, 8AK, UK
4 International Institute of Agri-Food Security, Curtin Business School, Curtin University, Perth, WA 6845, Australia

(Received 4 September 2014; Accepted in revised form 15 December 2014)

Summary Puff pastry is a popular nonleavened pastry product with unique textural properties. Puff pastry is processed from laminated dough, comprising alternating and discrete layers of dough and fat. Puff pastry quality can vary substantially depending on the raw material quality and production processes. The high fat content of commercial puff pastry usually containing significant amounts of saturated and *trans* fats, which are known to be associated with potential health risks, has led to increased demand for lower fat baked products. This review discusses the puff pastry production process and factors affecting puff pastry quality together with strategies to reduce total fat content and improving the fat composition in puff pastries.

Keywords Laminated dough, puff pastry, reduced-fat puff pastry, saturated fat, *trans* fat.

Introduction

Puff pastry is a nonyeasted, distinctly layered pastry with a flaky texture, which is featured in a wide range of products such as pie tops, sausage rolls, cream horns and vol-au-vent cases. Puff pastry is a laminated bakery product made from a paste consisting of many thin layers of dough separated (laminated) by alternate fat layers. Puff pastry is produced in different scale bakeries using semi-automated in small to medium scale, or fully automated in large scale mass puff pastry production lines (Bent, 1997). Puff pastry making is a complicated process that involves widely varying production practices and the making of puff pastry still remains an art.

As the production process depends on the quantity of fat, bakers typically use a fat content varying from 50% to 100% on flour basis in the recipes and the finished puff pastry can contain 30% or more fat on a weight basis (Simovic *et al.*, 2009), and hence, it is considered to be a high fat food. In addition, most of the fats used commercially may contain up to 50% saturated fat and 25–30% *trans* fat (Simovic *et al.*, 2009) which are considered by health agencies to be the least desirable fat types. Numerous clinical studies have demonstrated that excess consumption of saturated fat is positively correlated with elevated blood cholesterol, which is a major risk factor for cardiovascular disease. Furthermore, *trans* fats are worse than saturated fats and are associated with increased risks of a number of diseases, including coronary heart disease, cancer, diabetes, allergy and poor foetal development (Stender & Dyerberg, 2004; Wijesundera *et al.*, 2007). Producing foods with low saturated fat and free of *trans* fats has become a global health objective (Wassell & Young, 2007). Increased consumer concern over the quantity and quality of fat in pastry products has increased pressure on the bakery industry to produce low-fat baked products with a more desirable fat composition (Wassell & Young, 2007).

Croissant and Danish pastries are also manufactured from laminated pastes although in these products yeast is added to the dough and a fermentation period (proofing) is used to increase the product volume before baking (Bent, 2007). While there are recipe, processing and leavening mechanisms between puff pastry, croissant and Danish pastry, many of the considerations which might be applied to reduce fat levels in puff pastry will equally apply to croissant and Danish pastry.

In 1975, McGill summarised the basic production processes used in puff pastry in his review article which has not been updated. The present review aims to provide a summary of the puff pastry production process and a comprehensive overview of the factors affecting puff pastry quality, together with methods to reduce

*Correspondent: Email: mike.sissons@dpi.nsw.gov.au

doi:10.1111/ijfs.12754

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the fat content and reduce the level of saturated and *trans* fat in puff pastry.

Puff pastry production process and physicochemical changes

Dough and paste formulations

A basic puff pastry formula contains flour, salt, water and fat (Hay, 1993). Croissant and Danish pastry recipes include yeast, sugar, eggs and other 'enriching agents' (Bent, 2007). The fat used in laminated products has numerous terms such as 'laminating fat, 'roll-in fat' or 'layering fat'. Some formulae optionally use soft dough fats such as butter and margarine up to 10% on flour basis to improve handling properties or eating quality (Cauvain, 2001). Hereafter, the word '*dough*' used in the review refers to the mixture of ingredients (flour, water, salt) before incorporating the laminating fat, and the word '*paste*' refers to the mixture of dough and laminating fat before baking and the term '*pastry*' to the final baked product.

Dough mixing and laminating

In classical puff pastry production, flour, water and salt are mixed to obtain the base dough and the full development of the gluten structure is initially avoided as gluten development continues in the successive sheeting steps. After mixing the dough is allowed to rest before being extruded into a dough sheet. The laminating fat is extruded into a separate sheet which is then placed onto the formed dough sheet. The latter is then folded to enclose the laminating fat and the process of building up the separate fat layers sandwiched between dough layers commences. There are three commonly used methods that pastry cooks use for incorporating laminating fat into dough: English, French and Scotch methods for which the details are provided elsewhere (McGill, 1975; Bent, 1997; Harte, 2003).

In subsequent processing, the paste obtained is continuously sheeted and folded in a process commonly called 'laminating' or 'layering' the pastry (Patient, 1994). The purpose of folding is to form continuous alternate and ideally discrete layers of fat and dough. This is important to obtain the necessary lift and subsequent delicately layered baked pastry. The laminating is achieved by different folding methods, traditionally referred to as 'threefold turn' (half turn), 'fourfold turn' (bookfold) and 'singlefold turn' (McGill, 1975).

Between folding and sheeting steps, the paste is allowed to rest. This relaxation/retardation of paste is preferably done at ~4–8 °C to prevent temperature elevations in the paste and subsequent reduction in baking quality. The low-temperature retardation is more critical when butter instead of commercial puff pastry margarine is used as puff pastry fat. An overnight retardation of paste at ~4–8 °C is sometimes applied before the final sheeting step. The automated (continuous) production methods were summarised by Harte (2003) and Haarsgaard (1980).

Physicochemical changes that occur during the manufacture of puff pastry

During the mixing of flour with water, the proteins are developed into a continuous gluten matrix that binds starch granules, and subsequent processing ensures the required balance between cohesion, elasticity and extensibility of the dough. During baking, moisture in the paste turns to steam which cannot easily escape due to the impervious fat layers. The build-up of steam pressure forces the dough layers to separate from each other generating the lift of the pastry. In croissant and Danish pastes, the effect of steam pressure is supplemented by the release of carbon dioxide from yeast fermentation. The laminating fat is melted and absorbed by the dough giving it a soft texture and resistance to moisture gain which helps retain a crispness when wet fillings are used in the puff pastry. The gluten network formed throughout the pastry primarily determines whether the pastry is crumbly and tough or tender and flakey (Andrews & Harte, 2003). This is an oversimplified explanation which overlooks the complex physical and chemical transformations occurring in pastry baking (Kazier & Dyer, 1995).

Puff pastry fats

Level of fat in the formulation

Based on the laminating fat content used in the recipe, bakers express puff pastry as 'full-pastry' 'three-quarter pastry' or 'half-pastry' expressing the quantity of laminating fat relative to the flour weight (Cauvain, 2001). For example, the above-mentioned three different kinds of pastry have fat contents of 100%, 75% and 50% on flour weight basis, respectively. Nevertheless, a continuous range of recipes is possible using any per cent of laminating fat content in the range of 50–100% (McGill, 1975). The choice of level of fat to be used in puff pastry production by manufacturers depends on criteria expected in the final product such as lift, eating quality and flavour (Cauvain & Young, 2008).

Characteristics and types of puff pastry fats

Several researchers have described key performance characteristics of puff pastry fats: firmness (consistency),

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plasticity (optimal solid fat content) and melting properties (Tamstorf *et al.*, 1986; Kazier & Dyer, 1995; Cauvain, 2001; Wassell & Young, 2007; Pajin *et al.*, 2011; Lefébure *et al.*, 2013). The fat should be firm enough to withstand the dough folding operation and to form continuous undamaged thin layers of fat under a high degree of sheer applied during the sheeting. At the same time, it should be plastic enough to be workable and evenly spreadable, and flowable under pressure. The fat crystals have to be able to separate and move past one another in an oil phase during sheeting. The melting properties of the fat determine the crystalline solid fat content and the size of the crystals, which in turn influence the firmness and functionality of the fat (Cauvain, 2001). Generally, these rheological characteristics of puff pastry fat are achieved by the high melting point of saturated fats in the fat blend. However, melting points should be low enough at the same time to avoid a 'palate cling' (unpleasant waxy after taste) caused by unmelted solid fat in mouth (Tamstorf *et al.*, 1986). According to Cauvain (2001), fats that melt slowly between 15 and 20 °C, but rapidly between 25 and 45 °C, allow palate cling to be reduced without sacrificing plasticity.

A few studies have reported the optimum melting property requirement for a puff pastry fat. Doerry (as cited in Pajin *et al.*, 2011) indicated that a solid fat content of 16% at 40 °C and the requirement for gradual changes of this content at temperatures between 50 and 92 °C to be optimum. Similarly, a solid fat content of 38–45% at 20 °C reported to give the maximum-specific height of puff pastry (Cauvain, 2001).

The main types of laminating fat used for puff pastry making are butter, shortenings and margarines. Butter was traditionally the main fat used, but in many cases, it has been replaced by other fats, due mainly to its high cost and low-processing temperatures which may be required (Pajin *et al.*, 2011). For example, butter has a low solid fat index at 20 °C and puff pastry sheets made with butter need to be processed at temperature around 12–14 °C (Cauvain & Young, 2001). Nevertheless, butter is still being used today, especially for rich types of puff pastry because of its exceptional flavour profile (Gassenmeier & Schieberle, 1994; Kazier & Dyer, 1995). Rønholt *et al.* (2014) attempted to develop a 'puff pastry butter' using a scraped heat exchanger and changed the manufacturing conditions with the objective of reducing structural breakdown of butter during working.

Shortenings composed of 100% fat produced from vegetable oil or blends of vegetable and animal fats are widely used in pastry making (Kazier & Dyer, 1995), although there has been a move away from the use of animal fats. The melting profiles of these shortenings, combined with a considerable solid content at high temperatures, facilitate the plasticity and good workability required of fat during lamination and optimum expansion during baking. However, they often result in 'palate cling' due to the fact that their melting temperatures are above human body temperature.

Puff pastry margarines are designed specifically for pastry production and result in less palate cling. Puff pastry margarines are water-in-oil emulsions containing around 80% triglycerides, and their high content of solid fat allows melting over a wide temperature range. These margarines provide the high workability of high melting fats, but they have relatively lower melting points compared with shortenings (Kazier & Dyer, 1995). Puff pastry margarines have saturated fat (solid at room temperature) content of ~50%. The main saturated fatty acid is palmitic acid (C16:0) with lesser amounts of stearic acid (C18:0), tetradecanoic acid (C14:0) and dodecanoic acid (C12:0). The unsaturated fatty acids (liquid at room temperature) contain mainly oleic acid (C18:1), and lesser amounts of linoleic acid (C18:2), and α-linoleic acid (C18:3) (Pajin *et al.*, 2011).

Traditionally, the most widely used technique to produce solid fat from liquid oils has been partial hydrogenation; a catalytic hydrogenation of double bonds in unsaturated fatty acid chains of oils. Industrial hydrogenation of vegetable oils involves three phases (gas–solid–liquid) carried out in a batch autoclave with a nickel-based catalyst (0.01–0.15%) at 110–190 °C, 30–70 psi H_2 pressure. This process also allows the formation of *trans* isomers of unsaturated fatty acids. These *trans* isomers show different physiological properties and metabolic functions to their natural *cis* counterparts (de Cindio & Lupi, 2011). They provide the favourable melting behaviour, textural properties and oxidative stability to the fat. Due to the concern over health effects of *trans* fats in food, there is an increasing pressure to remove the term 'hydrogenated fat' from the food labels (Wassell & Young, 2007).

Factors affecting the quality of puff pastry

Although it is difficult to give a clear interpretation of what is meant by the quality of puff pastry, some of the quality criteria that make puff pastry acceptable to consumers have been identified and adopted (Table 1).

Puff pastry quality can vary substantially depending on the quality of the raw materials and variations in the production process (McGill, 1975; Hay, 1993; Cauvain, 2001). Indeed, changes in either of these factors may result in a sudden decline in quality and hence economic losses, especially in large scale production. The following section will discuss a few of these factors considering their relevant importance. This knowledge is vital in fat reduction as optimisation of these

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Table 1 Objective and subjective parameters used in the evaluation of puff pastry quality

Quality parameters	Details	Remarks	Reference
Objective parameters			
Pastry lift/ Specific lift	Pastry lift is the measurement of the height of the pastry $\text{Specific lift} = \frac{\text{pastry lift (mm)}}{\text{paste weight before baking (g)}}$	*Specific lift* provides a better method to compare pastry produced using different final paste thicknesses/weight than the lift. It is also an indicator of other parameters such as pastry firmness, shrinkage and volume	Gerrard *et al.*(2000), Hay (1993), McGill (1975), Simovic *et al.* (2009) and Sliwinski *et al.* (2004)
Development	$\text{Development} = \frac{\text{pastry lift (mm)}}{\text{height of the paste weight before baking (mm)}}$	Commonly used parameter by the bakers. Bakers' expectations of the development of puff pastry is 5–8.	Cavillot *et al.*(2009) and McGill (1975)
Pastry Volume/ Specific volume	Pastry volume is the measurement of volume of the $\text{Specific volume} = \frac{\text{pastry volume of the pastry (cm}^3\text{)}}{\text{paste weight before baking (g)}}$	Rape seed displacement method or laser-based scanners can be used for volume measurement.	Hay (1993) and Simovic *et al.*(2009)
Lift irregularity	Lift irregularity = maximum pastry height (mm) − minimum pastry height (mm)	Measures the aesthetic quality of pastry. The higher the unevenness of the lift the higher the irregularity value.	Telloke (1991)
Eccentricity	$\text{Eccentricity} = \frac{\text{(maximum top diameter of the pastry ring) - (minimum top diameter of the pastry ring)}}{\text{(minimum top diameter of the pastry ring)}} \times 100$	Higher values of eccentricity reflects more deformation of a perfect circular shape for pastry rings.	Telloke (1991)
Shrinkage	$\text{Shrinkage} = \frac{\text{(surface area of the pastry top)-(surface area of the pastry cutter)}}{\text{(surface area of the pastry cutter)}} \times 100$	Measures product contraction during baking and aesthetic quality of pastry. There is a direct relationship between pastry height and shrinkage. Higher shrinkage values mean higher product deformation. Pastry from stronger flours has more pastry deformation.	Telloke (1991)
Instrumental firmness	Different compression tests can be used to evaluate the firmness, defined as maximum resistance to force. Ex: a blade or probe attached to the texture analyser forcing through pastry to obtain maximum shear load	Telloke (1991) found the firmness measurements agree with panel assessment of pastry tenderness ($r^2 = 0.96$).	Simovic *et al.* (2009) and Telloke (1991)
Structure	Image analysis is used to analyse the image of pastry cross-sections for different parameters such as cell size and cell wall thickness	Novel method. Salmon (2005) analysed the air cell distribution in a pastry cross-section and attempted to relate to subjective quality.	Salmon (2005)

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Table 1 (*Continued*)

Quality parameters	Details	Remarks	Reference
Subjective parameters			
External appearance	Crust colour (yellowness), colour uniformity, surface glossiness, puffiness	Trained panellists are needed	Pimdit *et al.* (2008) and Simovic *et al.* (2009)
Structure	Crispness, flakiness, flakiness uniformity, pore size, pore wall thickness		Baardseth *et al.* (1995)
Flavour	Margarine, wheat		
After taste	Palate cling, saltiness, tooth pack, mouth dryness		

factors may compensate for the quality loss caused by fat reduction.

Flour quality

Bakers commonly use flour protein content as the primary criterion for selecting the type of flour to be used for puff pastry (Hay, 1993) although this differs between countries. For example, a protein content of 12–14.5% is specified for Europe while 8–10% is optimal in New Zealand. It is generally accepted that strong flours give increased lift (Taylor, 1985) and weak flours have low lift which could be due to destruction of the layers early in lamination, resulting in poor gas retention and low pastry lift. Many researchers found that the rheological quality of the gluten protein during baking and processing is of greater importance than the protein content of the flour.

Hay (1993), using eighteen different wheat flours, reported pastry-specific height was strongly influenced by water absorption and rheological properties of the dough particularly extensigraph tension energy. The study revealed the quantity of high molecular weight glutenin subunits was strongly correlated with extensibility ($r = 0.85$). Extensibility is desirable to enable the paste to withstand sheeting and avoid rupturing of the laminates and tearing of the dough sheets. Additionally, it was observed that the bread baking quality of flour does not provide a good indication of pastry baking quality of the same flour.

Sliwinski *et al.* (2004) using eight wheat cultivars found that puff pastry baking quality depended on a few stress-related dough properties such as strain hardening ($r = 0.95$). A high strain hardening co-efficient helps to prevent premature rupture of dough films to ensure uniformly thin dough layers and increased the pastry lift. They also found no significant correlation ($r = 0.32$) between pastry volume and gluten content of the flour. There was a positive correlation between pastry volume and optimum mixing time required for bread production ($r = 0.84$).

Ludewig (1978) as cited by Telloke (1991) proposed an optimum Extensigraph dough resistance of 300–500 Brabender Units. However, the validity of using the methods developed for assessing bread baking quality of flour for determining the pastry baking quality is questionable as pastry dough is quite different to bread dough in composition and rheological changes during sheeting (Telloke, 1991; Morgenstern *et al.*, 1996). Therefore, it is important to develop methods that are more applicable to measure the baking quality of flour for puff pastry. Telloke (1991) measured 'dough deformation resistance' of a dough sheet; the force required for the standard deformation of a dough sheet obtained after a standardised sheeting procedure. This

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was measured using a texture analyser with spherical probe to deform the tightly held dough sheet.

Dough water content

Achieving optimum dough consistency and reduced dough stickiness is crucial in puff pastry production (Sliwinski *et al.*, 2004). The amount of water used depends on the flour water absorption, the amount of dough fat and the paste making process (McGill, 1975). It is generally believed that a higher water level in paste produces higher pastry lift. However, Telloke (1991) found that pastry-specific height and shrinkage were hardly affected when dough water content was increased from 52.5% to 62.5% (flour weight) while keeping the flour to pastry margarine ratio constant. Whereas Sliwinski *et al.* (2004) reported increased dough water content from optimum reduced pastry-specific height. Differences in wheat cultivars used and the increased water level might have increased dough stickiness causing amalgamation of dough layers and reducing the expansion of them in latter work.

Fat content

An increase in the laminating fat content increases the specific lift of the baked puff pastry (Telloke, 1991). This is probably due to the thicker fat layers produced by the high fat content and minimal sticking together of dough layers during lamination. Further, it can provide a greater impervious barrier to the escape of steam in the early stage of baking. Reducing the laminating fat content below 50% without making other recipe or process changes will inevitably result in the reduction of pastry lift and other eating qualities (Kazier & Dyer, 1995) unless other steps are taken to preserve the characteristic structure. Nevertheless, frozen puff pastry paste sheets having laminating fat content below 50% on flour basis are available in the market.

Degree of lamination

Degree of lamination can be expressed as the number of theoretical fat layers in the final paste, which is calculated from the initial number of fat layers and number of folds given subsequently (Cauvain, 2002). The optimum level of lamination varies with the amount of laminating fat used in the recipe (Cauvain, 2001). The optimum degree of lamination shifts depends on the fat content. For example, 128 fat layers were found to be optimum with half paste (50% of fat of flour weight) and 256 fat layers gave best quality pastry with a full paste (100% of fat of flour weight).

Resting of dough and paste

The dough and the laminated pastes are commonly given different resting periods for stress relaxation. Inadequate relaxation can cause undesirable shrinkage and tearing of the dough during lamination (Morgenstern *et al.*, 1996; Bousquieres *et al.*, 2014). The toughness and resistance to stretching of wheat dough reduces with time after mixing, reflecting a gradual change in the gluten structure with a gradual reduction in elasticity. During relaxation, the dough attains an optimum rheological state, but this occurs at different rates for flours of varying strengths and relaxation times. Cauvain (2002) found that doughs made of weak flour need little or no relaxation, while doughs of stronger flour need longer relaxation periods.

Baking temperature

Puff pastry should be baked in a hot oven to ensure fast steam generation which expands the pastry during the initial stage of baking. The baking temperature varies according to the type of puff pastry (McGill, 1975). For example, products with filling need a temperature between 216 and 221 °C, while plain unfilled pastry needs 227–238 °C. Lower temperatures are generally used for a convection oven to counteract premature crust solidification and low pastry lift caused by heat flux in a convection oven (Telloke, 1991). The baking process can be optimised by increasing the baking temperature during the early stage of baking when pastry expansion occurs then reducing the temperature for drying the pastry without burning (Telloke, 1991).

Refrigerated and frozen storage of finished paste

In common practice, preprepared puff paste pastes are kept refrigerated. Refrigerating at temperatures below 5 °C provides superior protection against quality deterioration with higher pastry lift than storage at 10–15 °C (Leung *et al.*, 1984; Cauvain, 2002). Dark spot formation is another form of quality deterioration occurring during prolonged cold storage (Cauvain & Young, 2008) which is attributed to the enzyme-assisted oxidation of polyphenols in the small bran particles. This discoloration can be reduced by addition of ascorbic and citric acid to the dough (Cauvain, 2001) although this may affect pastry shrinkage.

Frozen puff pastry paste sheets and bulk slabs are becoming increasingly popular in the market by consumers and retail outlets. These frozen products are stable having a shelf life of 6–12 months due to their high fat content. However, Gerrard *et al.* (2000) reported an 18.5% loss of puff pastry lift during frozen storage at −20 °C over 3 months possibly due to disruption of the gluten network by ice crystals and loss

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of dough strength (Inoue *et al*., 1994). Nevertheless, Gerrard *et al*. (2000) did not investigate the effect of frozen storage on rheological properties of paste.

There have been a few scientific attempts to improve low-temperature storage stability and prolong the storage period of puff pastry dough. Leung *et al*. (1984) showed that reducing the water activity of puff pastry paste from 0.98 to 0.90 by including sorbitol in the recipe improved the storability at 5 °C. Gerrard *et al*. (2000) showed that adding microbial transglutaminase, (an oxidising flour improver) at 5000 ppm (based on dry flour weight), improved pastry lift and could sustain it through frozen storage at -20 °C for up to 3 months. This might be due to the oxidation of cysteine residues of gluten protein by microbial transglutaminase facilitating disulphide bonds and cross-linking, hence strengthening the protein matrix against the deleterious effects of frozen storage.

Storage of baked pastry

Puff pastries have a limited shelf life of around 7 days because of their hygroscopic nature and they become tough and chewy in the staling process as they absorb moisture from the surroundings (Ke & Brown, 1985). This problem is aggravated in pastries with moist fillings, as the filling moisture migrates to the pastry shell (Van Der Graaf & Verhoef, 1995). In their patent, Ke & Brown (1985) claimed a modified puff pastry product with longer shelf life and less sogginess compared with conventional puff pastries. Wheat flour was partially substituted with dextrin (preferably 20–25%). Less 'sogginess' was observed in the modified puff pastry than the conventional puff pastry.

Opportunities for reducing total fat, saturated and *trans* fat content in puff pastry

Reduced-fat pastry

There are several approaches available for reducing the fat in laminated products: (1) replacing the laminating fat (2) reducing the amount of laminating media (3) reducing the fat content of the laminating media and (4) using emulsifiers.

(1) Replacing the laminating fat is simply replacing part of the laminating fat with a fat replacer. These fat replacers are either based on carbohydrate, protein or lipid or a mixture of them. Before selecting a fat replacer, it is important to understand the role fat plays in the food and select fat replacer(s) based on this. For puff pastry, the most important characteristic of a fat replacer should be its ability to form hydrophobic impervious layers. This suggests suitability of lipid-based fat replacers over the other categories. Carbohydrate-based fat replacers widely used in other bakery products can be less effective in obtaining flakiness because of their hydrophilic properties (Kazier & Dyer, 1995).

Lipid-based fat replacers are fat analogues that cannot be metabolised by digestive enzymes and are therefore lower in calories. However, there is public concern over the gastrointestinal safety of fat replacers despite the rigorous safety evaluation programmes they undergo for approval as a food additive. This is because of potential adverse effect of unabsorbed fat on digestion and absorption of nutrients and gastrointestinal lymphatic tissues and bile acids (Munro, 1990).

Pimdit *et al*. (2008) used carbohydrate (maltodextrin gel and powdered cellulose), protein (whey protein 80%) and lipid (Salatrim)-based fat replacers to develop a puff pastry having a fat content of 12.88–15.86%. Salatrim, the acronym for 'short and long chain acyl triglyceride molecule', is a reduced-calorie fat substitute. Although the different fat replacers did not affect pastry-specific volume, their sensory appearance (crust colour, glossiness, colour uniformity, surface burning and puffiness) and texture (crispness, firmness) were generally found to be significantly lower than that of regular-fat puff pastry and Salatrim gave an unpleasant rubber odour. However, regular-fat pastry gave more oily and margarine flavour and waxy after taste than the reduced-fat counterpart which might be perceived as undesirable by health conscious customers.

(2) Reducing the amount of laminating media can reduce lift directly but can be used for products that do not require a high degree of flakiness. However, if a reduction in laminating fat level is combined with changes in laminating procedures (i.e. reducing number of fat layers), then lower levels of fat in the final product can be achieved.

(3) Reducing the amount of fat in the laminating media has been achieved for reduced-fat Danish pastry and croissants by developing special laminating media claimed to have only 50% fat and 44% water and functional ingredients (apparently emulsifiers and thickening agents) in its composition (Kazier & Dyer, 1995). However the study did not investigate the effect of fat reduction on the objective and subjective quality attributes of these pastries.

(4) Emulsifiers are a form of fat replacer and are commonly referred to by the baking industry as fat extenders. Puff pastry margarines are often incorporated with emulsifiers such as lecithins (Bueschelberger, 2004), mono- and diglycerides (Moonen & Bas, 2004) and polyglycerol esters (Norn, 2004). Emulsifiers play a significant role in the structure and plasticity of the margarines which facilitate pastry making. An emulsifier stabilises the margarine (oil–water emulsion) and aids in dispersion of the small fat crystals in baked products. There is evidence (S.P. Cauvain, personal

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communications) to show that small and uniformly dispersed fat crystals can improve the functionality of puff pastry margarines when overall fat levels are reduced. Puff pastry recipes generally do not include emulsifiers. However, ionic emulsifiers such as DATEM (diacetyl tartaric acid ester of mono- and diglycerides) and SSL (sodium stearoyl-2-lactylate) at 1–2% of flour weight were found to increase pastry lift and tenderness (Cauvain, 2001). Interestingly, it was observed that at reduced levels of laminating fat, responses to DATEM addition were more pronounced. The increased lift of pastry by these anionic emulsifiers was attributed to decreased dough pH, which was associated with an increase in paste resistance to deformation. Nonetheless, the effect of emulsifiers on pastry tenderness was not pH dependent. Cauvain (2001) observed that the nonionic emulsifier glycerol mono stearate did not affect pastry lift, but produced a more tender pastry. Simovic *et al*. (2009) used an ionic emulsifier, DATEM in puff pastry formulation using low *trans* fat margarine apparently to overcome the reduction in the puff pastry lift, due to the use of *trans* fat-free puff pastry margarine.

Production of margarines with lower levels of *trans* and saturated fat

As discussed earlier in this review, fats conventionally used in puff pastry production are high in saturated and *trans* fat and replacing them presents a challenge because of their functional role in pastry making. Regulating the commercial hydrogenation process to obtain low *trans* fat content is one option. For example, increasing the pressure, and agitation or decreasing the temperature or using precious metal catalyst such as Pd, Rh, Pt and Ru in the hydrogenation process helps to produce less *trans* fats (Singh *et al*., 2009; de Cindio & Lupi, 2011).

An alternative process that does not produce *trans* fat is interesterification between fully hydrogenated fat and *trans* free natural vegetable oil. The interesterification process catalysed by enzyme (preferably) or chemicals involves exchange of fatty acids between participating glycerol molecules and rearrangement of molecules (Lee *et al*., 2008). Saturated fatty acid chains in fully hydrogenated fats can be replaced with unsaturated chains of triglycerol naturally present in vegetable oil. The new altered triglycerol molecules give desired physicochemical properties to the restructured fat.

Palm stearin, a solid fraction of palm oil, is known to promote desirable plastic fats economically when it is interesterified with vegetable oils (Wassell & Young, 2007). Another option is adding hard fats to the liquid oil phase with suitable emulsifiers and stabilising agents to obtain blends. Previous studies investigated producing margarines with improved composition based on liquid oils such as olive oil using phospholipid stabilisers and mixing with hard fat fractions such as cottonseed stearin, palm and canola stearin and intermediate melting point triglycerides such as interesterified fat, cocoa butter and/or fractionated fat with melting point of 30–40 °C (de Cindio & Lupi, 2011).

Novel fat blends low in *trans* fat

Low *trans* fat margarine targeted for puff pastry production has appeared in the European market (Pajin *et al*., 2011). A high percentage of palm stearin (80–90%) is used in the production of a typical low *trans* margarine to obtain the required functionality (Wassell & Young, 2007), but these are high in saturated fat. Pajin *et al*. (2011) compared low *trans* fat (<1%), medium *trans* fat (5.4%) and high *trans* fat (23.8%) margarines for their physico-chemical properties. The *trans* fat content of the low and medium *trans* margarines had a higher saturated fat content (50–52%) compared with conventional puff pastry margarines (~30%). Furthermore, they found the ratio of *cis* polyunsaturated fatty acid to saturated fatty acid for these novel margarines was lower than the 0.50 which is the recommended optimum level by the World Health Organization. This emphasised that the *trans* fat content alone is not adequate in assessing the nutritional quality of low *trans* fat formulations.

Simovic *et al*. (2009) reported the success of producing puff pastry with satisfactory quality using two different commercial low *trans* margarines at a fat content of 35% for one and 55% for the other. However, they did not compare the baking performance with those of commercial high *trans* margarines. Moreover, the satisfactory lift they obtained may have been due to the higher paste thickness (10 mm) used than the thickness used in regular practice (~4 mm). The two different margarines differed in their firmness despite having similar fatty acid composition which in turn resulted in different optimum fat levels to be used. They speculated that the firmness differences of the margarines were due to the differing distribution of fatty acids in the triglycerides. A similar study comparing *trans* free margarine with *trans* fat ones found different physical properties such as texture and plasticity, and this influenced baking performance (Cavillot *et al*., 2009).

Skogerson *et al*. (2010) demonstrated that high diglyceride emulsifiers facilitated the formulation of *trans* free puff pastry margarines from nonhydrogenated liquid vegetable oils in their patent which claims that 'nonhydrogenated vegetable oil-based margarine for puff pastry containing an elevated diglyceride emulsifier' (Skogerson *et al*., 2010). The emulsifiers in

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the claim have a higher content of diglycerides (preferably 70–80%) (Skogerson *et al*., 2010). The novel puff pastry margarine is prepared by admixing the high diglyceride emulsifiers (10–16% on w/w basis), nonhydrogenated vegetable oil (55–65%) and saturated fat (13–25%). They used a blend of refined palm oil and soybean oil as the nonhydrogenated oil base. Nutritional analysis of the margarine revealed that the sum of saturated fat plus *trans* fat for these novel margarines was lower than that of commercial low *trans* margarines. The new margarines gave comparable puff pastry lift to that of standard commercial products with partially hydrogenated fat.

Novel fat blends high in unsaturated fat

The expertise on using fat sources with unsaturated fat in pastry is limited compared with that of saturated fat (de Cindio & Lupi, 2011). There are few reports on developing puff pastry margarine from fat blends with high unsaturated fat content. Gabriele *et al*. (2008) attempted to develop an olive oil-based emulsion having similar properties to the industrial puff pastry fats, targeting frozen puff pastry production. The emulsion was formulated using olive oil, emulsifier and hydrophilic thickener agents (flour based). The large differences in rheological properties observed between the novel formulation and commercial margarine indicated the need for further improvement of novel fat.

Palm stearin, obtained through fractionation of palm oil, can be used to substitute the hydrogenated oils, rich in *trans* fatty acids (Aini & Miskandar, 2007). Garcia-Macias *et al*. (2011) formulated different fat blends using palm fractions (palm stearin, palm mid fraction and interesterified palm mid fraction) in combination with sunflower oil. These blends had low saturated fatty acid content (29.2 ± 0.85%), less than that of butter (50%) while having higher solid fat content than butter. The functionality of these blends as puff pastry fat was investigated in a warm processing environment (20–24 °C). The specific volumes of the pastry from these novel blends were significantly lower than that of butter despite the fact that the latter was used above its optimum processing temperature (typically <20 °C). They found a blend having palm stearin: palm mid fraction: high oleic sunflower oil of 13:19:68 gave pastry with similar instrumental texture (compressibility) to that of pastry from butter while the other blends tested gave hard pastry. de Cindio & Lupi (2011) proposed two potential colloidal interactions that might be related to producing novel fats blends: 'droplet attractive interaction' (small water droplets dispersed in nematic liquid crystals) and 'formation of interpenetrating polymer network' (Poulin *et al*., 1997; de Cindio & Lupi, 2011).

Implication of fat reduction and modification of fat composition on puff pastry

As already stated above, production of puff pastry with reduced fat, which still possess suitable functionality that the fat provides, posses a great technical challenge for food technologists and bakers. Reduced-fat puff pastry lacks pastry lift, flakiness, crispness and has higher dryness and hence firmer eating qualities than regular puff pastry (Sharp, 2001). A moist filling often used with puff pastry can increase the sogginess of the reduced-fat pastry because with less fat the moisture barrier properties of the pastry crust are reduced. Reduced fat paste can impose handling difficulties for the bakers as it can be firmer, tougher and/or sticker than full-fat paste. Although the reduced level of fat facilitate less waxy after taste, which is considered desirable, it can affect the flavour and aroma chemistry undesirably.

Reducing saturated fat and *trans* fat in puff pastry margarine reduces the plasticity of the margarine. The low-melting unsaturated fat of these margarines can melt and absorb into the dough quickly due to dough warm up during sheeting. As a result, adjacent layers can stick together and the layering effect will be lost. Melting of soft fat into oil can be exaggerated in croissants and Danish pastry which are proofed at higher temperatures (~35 °C) prior to baking. Moreover, the low saturated margarines will melt rapidly during early baking without facilitating the lift. This leads to loss of lift and flakiness. The fat blends low in saturation (~30%) generally produce firmer dough and pastry compared with butter (Garcia-Macias *et al*., 2011). However, the effect of these novel fat blends on flavour and aroma and overall sensory quality has not been studied.

There is considerable evidence on the effect of the triglyceride composition on flavour development, flavour release and aroma delivery of different foods, but this information is lacking in bakery products. Baardseth *et al*. (1995) using eleven different laminating fats made of milk fat or vegetable oil for Danish pastry production found the type of shortening influenced the taste, odour and colour using principle component analysis. On the other hand laminating fat content in the formulae influenced the texture and colour. To our knowledge, no studies have been reported in the effect of fat reduction in bakery products on flavour.

Conclusion

Saturated fat and *trans* fat reduction in puff pastry and other pastry products has become a pressing issue. A better understanding of the role of fat in the structure, flavour and colour development of pastry as well

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as the interactive role of fat and other ingredients is needed. With the increased interest to develop low-fat puff pastry and other bakery products, there is a need for identifying cost-effective novel fat replacers. Individual fat replacers have been previously employed in scientific work to reduce the fat content in puff pastry. These attempts reported poor success when a single fat replacer was used to achieve all the functions of fat in puff pastry.

Another research gap is the potential for making a nutritionally improved pastry composition through the use of whole grain flours. However, the presence of bran disrupts the integrity of the dough layers and commonly leads to loss of lift and flakiness. Whole grain flours do not reduce fat levels *per se* but contribute to increased fibre content.

There are no simple options for making reduced fat and/or *trans* fat laminated products because of the unique role that the fat plays in the formation of the structure of paste products. Clearly to deliver reduced-fat products will require an integrated product development approach which combines the options for different fat replacers and/or other additives, changes in the laminating compositions and adjustment of the production process. In this context, laminated products are no different than almost all other baked products in that a better understanding of the contribution of dough production and processing will make significant contributions to achieving new developments. As yet, no significant attention has been paid in existing studies on reduced-fat laminated products to the key role that processing may play.

Acknowledgments

The authors are indebted to the University of Adelaide for the financial support provided for a research project on the effect of novel wheat blends on puff pastry quality and for a postgraduate scholarship. The authors are also grateful to Dr. Colin Jenner and Dr. Beverly S. Muhlhausler for reading the draft. Further, we would like to acknowledge Mr. Fee Lee and Mr. Brian Lavery (Bakery Studies, TAFE, SA) for supplementing the practical knowledge on puff pastry production.

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